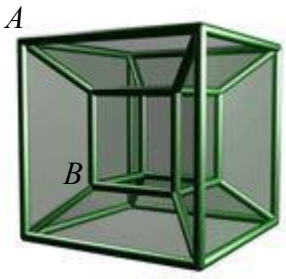
I. Fill in the blanks (5 points each, 20 in total).

1. Turn $y = f(x)$ once about the vertical y axis. A ring has free radius a, mass m、and stiffness k. The surface is smooth; the ring is elastic, circular and horizontal. For it to remain at rest at any height, the curve equation is _____.

2. The frame is a 3D projection of a 4D hypercube. It consists of inner and outer cubes with corresponding vertices linked. If every edge is made of material of resistance R, the equivalent resistance across AB is _____.



3. A bee sees light of mean wavelength 500nm. Its eye has about 5000 ommatidia. They densely cover the whole compound eye. Each intricate unit has a clear round collector (corneal lens) above a clear conical crystalline body. Light focuses on photosensitive cells at the cone tip and cannot cross its walls to another unit. Each makes one image point (pixel); these form a complete image.
Model the compound eye as a spherical cone with spherical radius r of 1.5mm. Then the optimum lens diameter d is about **(give two significant figures)** _____.

4. Open-circuit voltage U_{O} and short-circuit current I_{SC} are key for a p-n solar cell.

For this semiconductor, U_{O} = _____; the symbols denote respectively _____.

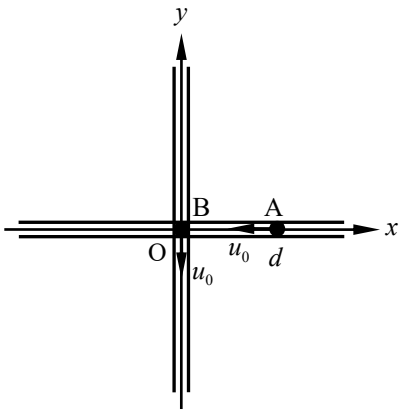
II. (15 points) Computer simulations display astronomical or microscopic motion by scaling geometrical dimensions. For the displayed motion to be physically possible in the scaled system, time must also be scaled.

1. A Newtonian particle orbits in a force field $\boldsymbol{F}(\boldsymbol{r})$ satisfying $\boldsymbol{F}(\alpha\boldsymbol{r}) = \alpha^k \boldsymbol{F}(\boldsymbol{r})$,

k is constant; $\boldsymbol{r}$ is the position vector. For dimension scale α, find the time scale β.

2. Hence show that a planet's orbital period squared scales as its orbit's size cubed.

III. (20 points) Two smooth thin tubes intersect at right angles in a level plane; each holds a particle of mass m and charge q.



For A and B, the initial separation of A from crossing O is d; B is at O. Their velocities are perpendicular, both $u_0 = \sqrt{\dfrac{kq^2}{md}}$, as shown; k is Coulomb's constant.

Find their minimum separation in the subsequent motion.

IV. (10 points) In thermodynamic cycles, engines take heat from a hot source, output part as work and reject the rest to a cold source. Heat pumps use external work to move cold-source heat to a hot source, together with that work converted to heat. By the second law, if either working substance contacts only two reservoirs at T_1、T_2, absorbed heats Q_1、Q_2 obey $\frac{Q_1}{T_1}+\frac{Q_2}{T_2}\le 0$. Positive and negative values mean heat absorption from, and heat rejection to, a reservoir, respectively.

A heater uses a boiler at T_0 to heat a room directly, maintaining a constant room temperature T_1, above outdoors at T_2. Using the same energy source and the machines, what is the theoretical maximum fractional energy saving compared with direct heating?

V. (15 points) Magnetic fields affect electrons, so the current-voltage relation may depart from the familiar Ohm's law. This problem studies one such example.

A 2D electron gas in plane xOy has areal density n (electrons per unit area). Along positive x is a uniform electric field $\boldsymbol{E} = E\boldsymbol{i}$ （$\boldsymbol{i}$ is the x unit vector); along z, the normal magnetic field is uniform, $\boldsymbol{B} = B\boldsymbol{k}$ （$\boldsymbol{k}$ is the z unit vector). The drag from scattering is proportional to velocity $\boldsymbol{v}$, written with time parameter τ as $\frac{-m\boldsymbol{v}}{\tau}$, m is the electron mass. Find the steady x and y current densities (current per unit transverse length) j_x and j_y using charge magnitude e、n、m、E、τ and ω, where $\omega = \frac{eB}{m}$.

VI. (25 points) In Fig. A, two thin parallel plates are ideal conductors, spaced *d*, infinite in *y*. An EM wave between them travels normal to *y*, along positive *x* as a travelling wave， with electric field

$E = E_0 \sin(2\pi z / \lambda_z)\sin(2\pi x / \lambda_x - \omega t)$. ω is

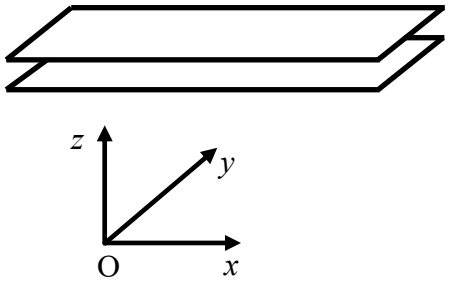


Fig. A

angular frequency; t is time, λ_z, λ_x are unknown.

Combining these makes practical microwave transmitting antennas, replacing huge conventional parabolic dishes and greatly reducing antenna cost.

1. Prove that λ_z can take only these values:

$$\lambda_z = 2d / m, \quad m = 1, 2, 3\ldots$$

2. For *m*=1, find λ_x .

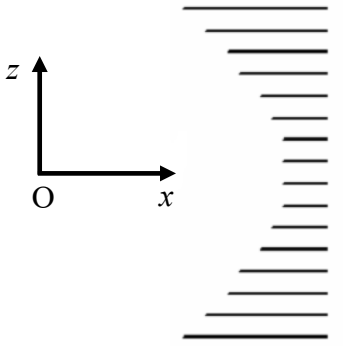


Fig. B

3. Stack equal-gap plates of unequal lengths, parallel and symmetric about an infinite *y* line source at O. Their right ends align; their lengths vary.
Fig. B is the cross-section in the page. At the right end, a plane EM wave is to emerge along *x*. Find the equation of the left boundary in the *x*O*z* section (Fig. B). (Take *m*=1; the horizontal distance from the source to the stack's left end is *L*）。

VII. (35 points) 1. Nuclear electric force makes an electron circle in classical hydrogen. Classical EM theory says accelerated electrons emit waves; the radiated power is given by (Larmor's formula) $P=\frac{e^2a^2}{6\pi c^3\varepsilon_0}$; a is acceleration, c vacuum light speed, $\varepsilon_0=\frac{1}{4\pi k}=8.854\times10^{-12}\,\mathrm{Fm^{-1}}$; the electron charge magnitude is $e=1.602\times10^{-19}\,\mathrm{C}$. Neglect relativity and estimate a classical atom's lifetime τ. (Measured binding energy is E_B=13.6eV, and electron rest mass is m_0=9.109×10^{-31}kg）

2. Radiation from accelerated charges has important uses. Synchrotron radiation, widely used in modern research, comes from near-light-speed electrons curving in magnetic fields. Its core is the electron storage ring; bunches traverse bending magnets and other devices to make high-quality synchrotron light. The recently built Shanghai facility is a third-generation source, with electron energy E=3.50GeV and bunch current I=300mA, circumference L=432m, cells (bending-magnet turns) N=20, and dipole field B=1.27T. Find mean total radiated power per outlet P_O.

(Near-light-speed electrons obey Newton's second law if relativistic mass is constant; the Larmor formula becomes $P=\frac{e^2a^2}{6\pi c^3\varepsilon_0}\times\gamma^4$, where $\gamma=\frac{E}{m_0c^2}$, E is total energy, and m_0c^2 is electron rest energy.)

3. Near-light-speed ring electrons emit synchrotron radiation in a cone along the orbit tangent, with half-angle $1/\gamma$. Thus higher electron energy gives better directionality.

For this facility, find the duration of a single electron's radiation received at a point along the radiation direction: ΔT.

(Give all results in this problem to three significant figures.)